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Evaluation of the sensitivity of the classical swine fever surveillance system in two free zones in Colombia

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Abstract

Infection with the classical swine fever virus (CSFV) causes a disease in pigs that ranges from a hyperacute form in which animals die in a few hours to subclinical disease. Due to this wide range of virulence, several complementary surveillance strategies should be implemented for the early detection of the disease. The objective of the present study was to determine the sensitivity of the surveillance system to detect CSFV outbreaks in a free zone (Zone 1) and in a zone undergoing an eradication process (Zone 2) in Colombia. Stochastic scenario tree models were used to describe the population and surveillance structures and to determine the probability of CSFV detection. The total sensitivity of the surveillance system in the case of a single infected farm in Zone 1 was 31.4% (CI 95%: 7.2–54.1) and in the case of 5 infected farms was 85.2% (CI 95%: 67.3–93.7), while in Zone 2 the sensitivities were 27.8% (CI 95%: 6.4–55.1) and 82.5% (CI 95%: 65–92.9), respectively. The on-farm passive surveillance shows the highest sensitivity for detection of a single CSFV infected farm in both zones (22.8% in Zone 1 and 22.5% in Zone 2). The probability of detection was higher in a family / backyard premise than on a commercial farm in both zones. The passive surveillance at slaughterhouse had a sensitivity of 5.3% and 4.5% for the detection of a single infected farm in Zone 1 and 2, respectively. Active surveillance presented a range of sensitivity between 2.2% and 4.5%. In conclusion, the sensitivity of the surveillance in the two studied zones was quite high, one of reasons for this good sensitivity being the sentinel network based on the voluntary participation of 5,500 collaborators that were trained for the identification and notification of diseases of national interest.

KEYWORDS

CSFV, scenario tree, stochastic modelling, surveillance

1 | INTRODUCTION

Classical swine fever (CSF) is one of the most important diseases in pigs, with major impact on animal health and on the swine industry (Blome et al., 2017; Penrith et al., 2011). It is caused by the CSF virus (CSFV), an enveloped RNA virus classified in the family *Flaviviridae*, genus *Pestivirus* (OIE, 2009; Ribbens et al., 2004). CSFV affects only

pigs and wild boars and can be transmitted both horizontally and vertically. CSFV infection can have a wide range of severity, from a chronic course to a hyperacute presentation in which animals die in a few hours. The clinical signs and lesions can vary among animals depending on the virulence of the CSFV strain and on such host factors as age, breed and immune status (Blome et al., 2017; Moennig et al., 2003; Petrov et al., 2014). This variability in disease

presentation forms and unspecific clinical signs means that the detection of the disease in free populations is not always easy and consequently, epidemiological surveillance should be based on different strategies. The CSF surveillance program should cover domestic and wild pig populations and should include an early warning system throughout the whole production chain, especially with regard to high-risk groups (OIE, 2019).

Active and passive surveillance refers to the method of data collection depending on whether the data provision is investigator-initiated (active) or observer-initiated (passive) (Dufour and Hendrikx, 2009).

Surveillance systems must be evaluated and optimized to improve their effectiveness regarding the data quality, costs and resources used. A good indicator to assess their effectiveness is the sensitivity (Se) of the surveillance system components, which has been defined as the probability of detection of the disease (or infection) if present at a certain level in the population. (FAO, 2014; Hoinville et al., 2013). This level is known as design prevalence and represents the 'threshold prevalence' from which the disease can be detected (Martin et al., 2007). The sensitivity of the surveillance depends on different factors such as the sensitivity (Se) and specificity (Sp) of the diagnostic test, on the number of animals included in the surveillance and on different circumstances related to activities performed (n).

CSF is still a major problem in some regions of Colombia. The country has been divided by CSF status into different epidemiological zones: a zone recognized as CSF free by the OIE, some zones undergoing an eradication process without vaccination and an affected zone where vaccination is compulsory (ICA, 2018). Despite the strict control of internal movements, the presence of the disease in the country supposes a major risk for the free zones. Surveillance is basic for achieving early detection of CSFV and the application of immediate measures to eradicate the infection (Donahue et al., 2012).

The objective of the present study was to determine the sensitivity of active and passive surveillance in the detection of CSFV infection in the two free zones in Colombia.

2 | MATERIALS AND METHODS

2.1 | Reference population

Colombia is located on the equator in the northwestern region of South America. With a swine population of 5.5 million head, different production systems coexist in the country from free-range backyard facilities to large industrial farms. CSFV is still present in a part of the country, specifically in the Atlantic Coast region where it is endemic, while the infection is absent in the other regions. However,

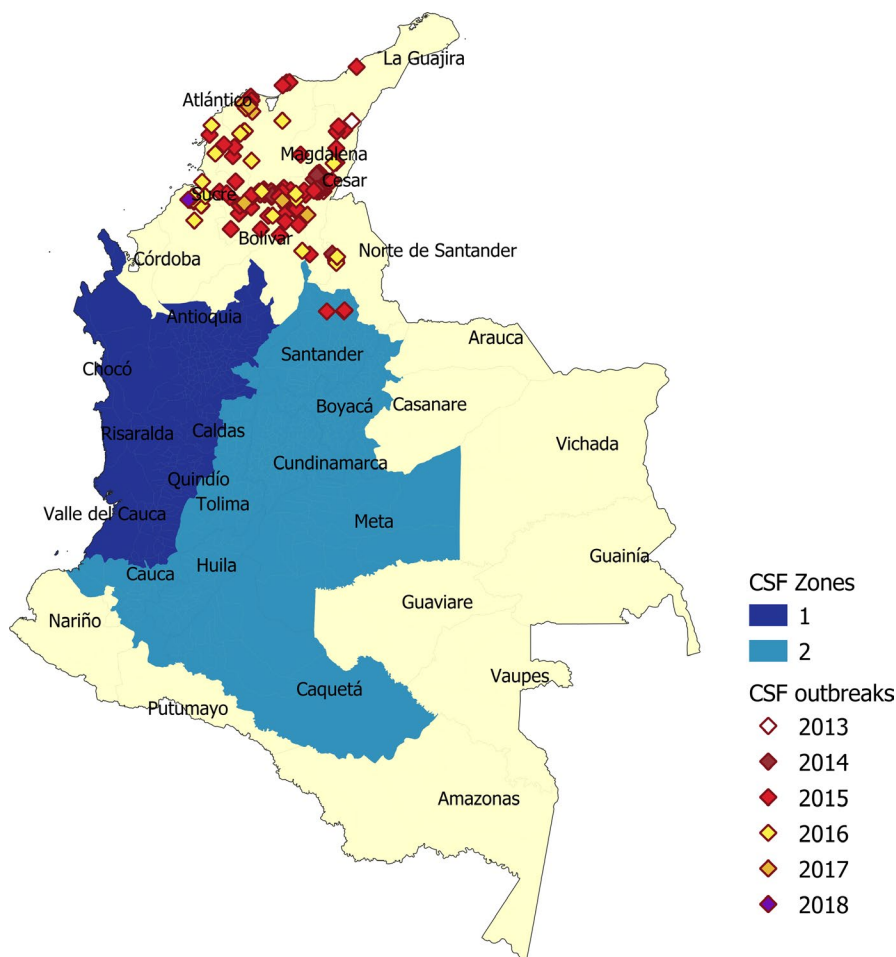


FIGURE 1 Map of Colombia with Zones 1 and 2 and CSF outbreaks occurring in the 2013–2018 period

the status differs: while in some regions vaccination is compulsory, in other zones it is prohibited. This paper focuses on two regions declared as free either by the government or by the OIE. (Figure 1). The free zone recognized by the OIE (Zone 1) includes 13,500 farms (15.5% of the total for Colombia), and the zone undergoing an eradication process (Zone 2) has 20,352 farms (25.6%). According to the database of the Colombian Association of Pig Farmers (Porkcolombia), 24% and 12.4%, respectively, are commercial farms, while the remainder is other family and backyard premises. (Table 1).

2.2 | The Colombian surveillance system

In Colombia, CSF is one of the notifiable diseases in the country and it is subject to official control, with active surveillance supporting disease control. Other than reporting by farmers and veterinarians, passive surveillance is based on a sentinel network created by the ICA (national veterinary services) as a component of its early warning system for diseases under surveillance (both endemic and exotic diseases). The purpose of the network is to increase the capacity to detect diseases in the country, and it is based on a participatory information system made up of 5,500 collaborators (farmers, technical assistants, veterinarians, farm managers and anyone else involved in the sector). People participate on a voluntary basis and have been trained by the ICA for the identification of clinical signs and reporting of diseases of national interest. The information is monitored, collected and analysed by the epidemiology section of the ICA. In zones 1 and 2, the network is made up of 1,247 and 2,386 members, respectively, who are mostly paraveterinarians, veterinary practitioners and animal identification handlers (people hired by Porkcolombia to individually identify pigs on farms) (ICA, 2019). Passive surveillance is also applied at slaughterhouses and is based on the detection of clinical signs of CSF in ante and postmortem inspection by veterinarians.

Epidemiological sampling for active surveillance of CSFV is performed yearly throughout the second semester. In this paper, we

evaluate the activities performed in 2018 and 2019 in zones 1 and 2. The parameters for sample size in 2018 and 2019 in Zone 1 were as follows: farm prevalence of 1% per each of the 5 categories into which farms are classified and an animal prevalence of 20% on commercial technified and genetic nucleus farms, 30% on commercial-industrial and family farms and 50% on backyard premises with a 95% confidence level. In Zone 2, the farm prevalence was fixed at 3% in 2018 and 1% in 2019 and animal prevalence was 20% in commercial technified, 30% on commercial-industrial and family farms and 50% for backyard premises. In the 2019 design, the same parameters were used, except for 1% farm prevalence and 60% animal prevalence on backyard premises. The number of farms and pigs sampled in both areas is shown in Table 1.

2.3 | Scenario tree model (STM)

The sensitivity of the surveillance system was calculated using a stochastic scenario tree model that describes the population and surveillance structures. The model represents all the elements involved in the different components of the surveillance network applied to the study population (FAO, 2014; Martin et al., 2007).

For each component of the surveillance system, the scenario tree defines a sequence of logical nodes with a different branch for each subpopulation considering the probabilities of infection and detection. The nodes are classified as infection, detection or risk nodes. The model is used to evaluate the probability of a randomly selected unit in the system giving a positive result.

Three design prevalences were set at 1, 5 or 10 affected herds, respectively, a single affected farm corresponded to a prevalence of 7.4×10^{-5} and 4.9×10^{-5} in zones 1 and 2, respectively. We assumed that the herd prevalence (P^*_H) and the within-herd prevalence (P^*_A) would be similar to those observed in the outbreaks of CSF detected on commercial farms and backyard premises in other regions of Colombia between 2013 and 2018 (Pineda et al., 2020). The detection nodes represented all the events or steps that had to occur to

TABLE 1 Number of premises and individual pigs sampled for CSF active surveillance in both zones between 2018 and 2019

Category of premises ^a	Zone 1					Zone 2				
	Number of farms	2018 Design		2019 Design		Number of farms	2018 Design		2019 Design	
		Sampled farms	No of pigs	Sampled farms	No of pigs		Sampled farms	No of pigs	Sampled farms	No of pigs
Genetic nucleus	18	18	270	17	255	4	4	60	3	45
Commercial technified	859	257	3,855	258	3,870	231	67	1,005	155	2,325
Commercial-industrial	2,365	66	660	91	910	2,292	49	490	66	660
Family	5,245	170	1,710	180	1,800	8,435	170	1,700	243	2,430
Backyard	5,013	240	1,200	252	1,260	9,390	206	1,030	310	1,240
Total	13,500	751	7,695	798	8,095	20,352	496	4,285	777	6,700

^aFarms grouped according to Government categories: Genetic nucleus (all should be tested), technified (farms with ≥ 100 sows and/or ≥ 600 fatteners), commercial-industrial (10–100 sows and/or 100–600 fatteners), family (3–10 sows and/or 15–100 fatteners) and backyard (≤ 3 sows and/or ≤ 15 fatteners).

detect the infection, while the risk nodes represented those factors that affected the probability of becoming infected or detected. The diagrams of the scenario trees for the three surveillance components are shown in Figure 2.

Table 2 shows the parameters included in the model. The relative risk for each category (commercial–family) was obtained by dividing the cumulative incidence on backyard and family farms by the incidence on commercial farms (reference category) observed in other regions of the country.

2.4 | Active surveillance

According to the endemic zone, small farms (backyard and family farms) have a higher risk of becoming infected than commercial farms. The relative risk (RR) of small farms compared with that of commercial ones and the proportion of both types of farm were used to calculate the adjusted risk of infection (AR) for both commercial and small farms in zones 1 and 2 (Equations 1 and 2).

$$AR_2 = \frac{R_2}{(R_1 * PrP_1) + (R_2 * PrP_2)} \quad (1)$$

$$AR_1 = AR_2 * R_1 \quad (2)$$

Where R_1 and R_2 were the backyard premise prevalence and commercial farm prevalence and PrP_1 and PrP_2 were the proportion of backyard premises–family farms and proportion of commercial farms, respectively. The sensitivity at farm level (SeHi), that is, the probability of detecting at least one positive animal, was calculated for each of the four categories of premises (Equation 3), which was given by the within-herd prevalence for each category i (P_A^*) multiplied by the sensitivity of the two tests (Se Elisa and Se PCR) applied in series and the number of tested animals (ni) on each farm according to the 2018 and 2019 samplings.

$$SeHi = 1 - (1 - P_A^* * Se\,Elisa * Se\,PCR)^{ni} \quad (3)$$

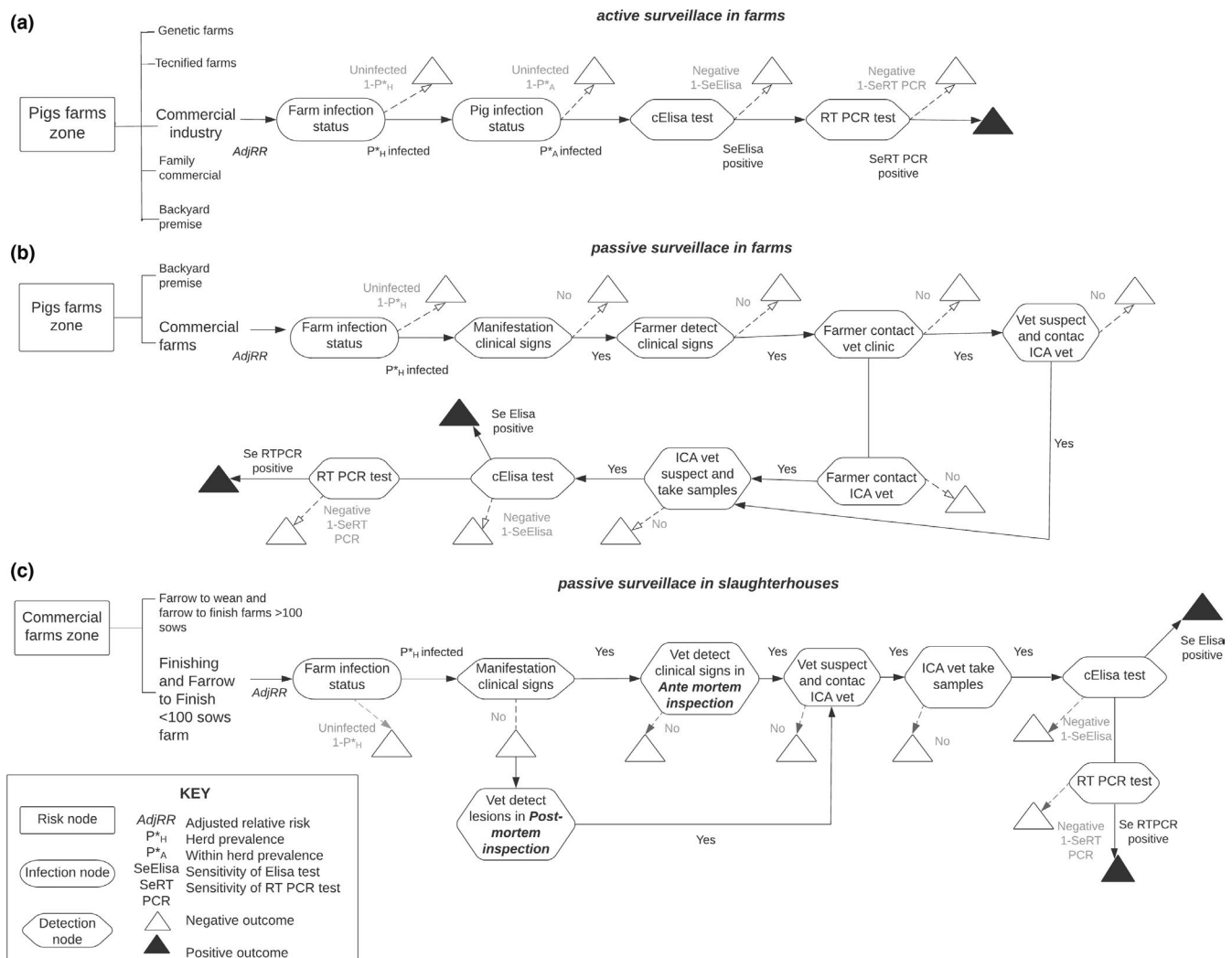


FIGURE 2 Scenario trees for active and on-farm and at slaughterhouse passive surveillance. Only the branch for commercial farms is represented, the others being of identical structure

TABLE 2 Variables and parameters included in the active and passive surveillance models

Node	Parameter	Abbreviation	Values and probability distributions in Zone 1	Values and probability distributions in Zone 2	Simulations distribution	References or sources of data
Variables and parameters common to the different paths						
Farm type	Number of commercial farms		3242	2527	Fixed value	Farms database - Porkcolombia
	Number of backyard premises and family farms		10258	17825	Fixed value	Farms database - Porkcolombia
Farm infection status	Proportion of commercial farms	PrP _c	999.9; 3164.5	999.9; 7058.5	Beta ^b	Farms database - Porkcolombia
	Proportion of backyard premises and family farms	PrP _b	999.9; 316.5	999.9; 142.4	Beta ^b	Farms database - Porkcolombia
	Design prevalence (for 1 infected farm)	P*	7.41E-05	4.91E-05	Fixed value	Zone 1: 1 /13500 farms; Zone 2: 1/20352 farms
	Farm-level prevalence (commercial) (R ₂)	P* _{Hc}	7.7; 8668		Beta ^b	7/8976 CSF outbreaks on commercial farms (Pineda et al., 2020)
	Farm-level prevalence (backyard) (R ₁)	P* _{Hb}	128.8; 123999		Beta ^b	127/123498 CSF outbreaks on backyard premises (Pineda et al., 2020)
	Relative risk for backyard premise	RR _b	1.318 (0.616 - 2.821)		Estimated value	Backyard premise prevalence / commercial farm prevalence Proportion with 95% CI according to (Morris & Gardner, 1988)
ELISA test	Adjusted risk for commercial farm (Reference)	AdjRR _c	0.805	0.782	Fixed value	$R_2 / (R_1 * PrPc) + (R_2 * PrPb)$
	Adjusted risk for backyard premises	AdjRR _b	1.062	1.031	Fixed value	AdjRR _c x R ₁ / R ₂
	Effective probability of infection of commercial farms	EPI _{Hc}	5.96E-05	3.84E-05	Fixed value	Adjusted risk by design prevalence
	Effective probability of infection on backyard premises	EPI _{Hb}	7.86E-05	5.07E-05	Fixed value	Adjusted risk by design prevalence
RT PCR test	Sensitivity of ELISA	Se Elisa	498.3; 27.2		Beta ^b	National Veterinary Diagnostic Laboratory - ICA
	Sensitivity of RT PCR	Se RTPCR	498.3; 27.2		Beta ^b	National Veterinary Diagnostic Laboratory - ICA
Manifestation of clinical signs	Probability of pigs on infected farm presenting clinical signs	CSI	0.65; 0.81; 0.95		Pert	Allepez et al 2007
Variables and parameters included in the Active surveillance on farm						
Pig infection status (intra-herd)	Animal level prevalence (commercial)	P* _{Ac}	550.5; 3812.6		Beta ^b	268/2130 diseased over susceptible pigs on commercial farms (Pineda et al., 2020)
	Animal level prevalence (backyard)	P* _{Ab}	999.9; 1020.2		Beta ^b	2617/5291 diseased over susceptible pigs on backyard premises (Pineda et al., 2020)

(Continues)

TABLE 2 (Continued)

Node	Parameter	Abbreviation	Values and probability distributions in Zone 1	Values and probability distributions in Zone 2	Simulations distribution	References or sources of data
Variables and parameters included in the Passive surveillance at farm						
Recognition of clinical signs by farmers and contact with veterinary services	Probability of commercial farmer calling the veterinary practitioner	F _C vet	0.80, 0.975, 1*	0.275, 0.40, 0.725*	Discrete	*25th, 50th and 75th quartiles.Quartiles obtained from a questionnaire issued to veterinary practitioners (discrete distributions in both zones included in the Supporting Information)
	Probability of commercial farmer calling the ICA	F _C ICA	0.00, 0.00, 0.05*	0.05, 0.10, 0.20*	Discrete	
	Probability of backyard farmer calling the veterinary practitioner	F _b vet	0.375, 0.65, 0.825*	0.187, 0.30, 0.425*	Discrete	
	Probability of backyard farmer calling the ICA	F _b ICA	0.015, 0.10, 0.105*	0.015, 0.20, 0.425*	Discrete	
Veterinarian contacts ICA	Probability of veterinary practitioner calling the ICA	VetICA	0.00, 0.00, 1*	1, 1, 1*	Discrete	*25th, 50th and 75th quartiles. Quartiles obtained from a questionnaire issued to veterinary practitioners (discrete distributions in both zones included in the Supporting Information)
Veterinarian takes samples	Probability of veterinarians performing a necropsy	VetNec	78.6; 44.2		Beta ^b	ICA: 86/134 necropsy in CSF outbreaks
	Veterinary services suspects CSF from signs and take samples	VetS _S	0.00, 0.50, 0.75*	0.5, 1, 1*	Discrete	*25th, 50th and 75th quartiles. Quartiles obtained from a questionnaire issued to veterinary practitioners (discrete distributions in both zones included in the Supporting Information)
	Veterinary services suspects CSF after necropsy and take samples	VetS _{NS}	0.75, 1,1*	0.687, 0.875, 1*	Discrete	
Variables and parameters included in the Passive surveillance at slaughterhouse ^a						
Farm type	Proportion of farrow-to-finish and fatteners on commercial farms	PrP _{fatt}	1000, 37.2	1000, 52.5	Beta ^b	Farms database - Porkcolombia
	Proportion of farrow-to-wean on commercial farms	PrP _{F-W}	975, 7883	730, 10626	Beta ^b	Farms database - Porkcolombia
Recognition of clinical signs by the slaughterhouse veterinarian	Antemortem inspection of fattener pigs	VetS _{AMIf}	0.00, 0.225, 0.70*	0.10, 0.50, 0.762*	Discrete	*25th, 50th and 75th quartiles.Quartiles obtained from a questionnaire issued to slaughterhouse veterinarians (discrete distributions in both zones included in the Supporting Information)
	Antemortem inspection of sows	VetS _{AMIs}	0.20, 0.50, 1*	0.275, 0.50, 1*	Discrete	
	Postmortem inspection of fattener pigs	VetS _{PMIf}	0.25, 0.50, 0.75*	0.437, 0.50, 0.75*	Discrete	
	Postmortem inspection of sows	VetS _{PMIs}	0.25, 0.25, 0.50*	0.437, 0.50, 0.75*	Discrete	
Slaughterhouse veterinarian contacts ICA	Veterinarian contacts ICA after suspicions in fattener pigs	VetICA _f	0.10, 0.50, 1*	0.45, 0.85, 1*	Discrete	
	Veterinarian contacts ICA after suspicions in sows	VetICA _s	0.20, 0.75, 1*	0.30, 0.60, 1*	Discrete	
	ICA veterinarian takes samples	ICAS	1			ICA:Samples are taken in all CSF report

^aOnly commercial farms have been taken into account at slaughterhouse.^bAll beta distributions are based on proportions with 95% CI according to clopper-pearson method (Branscum et al., 2005).

The probability of being infected depended on the type of farm (commercial versus backyard), which was defined as the effective probability of infection (EPIH_i) where *i* represented the type of farm and was given by the design prevalence (P^*) multiplied by the adjusted risk of each type of farm.

Finally, the sensitivity of active surveillance (Se_{active}) to detect an infected CSFV farm by serological sampling was estimated with Equation 4, where *i* corresponded to the type of farm and vector N_i is the number of farms sampled for each category farms. (Table 1).

$$Se_{active} = 1 - \prod_{i=1}^4 (1 - EPIH_i * SeH_i)^{N_i} \quad (4)$$

2.5 | Passive surveillance

Passive surveillance was applied both on farms and at slaughterhouses. The sensitivity at farm level was evaluated on commercial and small premises, while surveillance at slaughterhouses affected only commercial farms because animals from family and backyard production are generally slaughtered in uncontrolled facilities and are not sent to slaughterhouses.

When a private or slaughterhouse veterinarian suspects CSF, he/she had to contact the ICA. A veterinary officer examined the animals in the slaughterhouse to assess the condition and if necessary, visited the farm, collected samples and sent them to the official laboratory.

Figure 2 shows the scenario tree diagram with the main elements that determine the probability of detecting a positive farm through passive surveillance. A risk node divided the farms into two branches, backyard and family farms were included in the first and all other farms were grouped as commercial farms in the second. The surveillance unit was the farm, and the same design prevalences (P) as in the active surveillance model were used.

Due to lack of information, the expert opinion was used for the estimation of parameters of some probabilities using two questionnaires addressed at farm veterinarians (18 responses), farm paraveterinarians (9 responses) and slaughterhouse veterinarians (46). The questionnaires included two different clinical cases in sows and three in fatter pigs. Very shortly, in a farm with 50 breeding sows either: a) three sows aborted and two other pregnant sows died in 2–3 days; or b) eight piglets died and 30 piglets presented anorexia and weakness. In a fattening unit with 100 pigs: a) 20%; b) 40% of fattening pigs presented anorexia, fever and dyspnea in the last 2–3 days; c) mortality was 10% of animals in the last 10 days. Veterinarians answered about the probabilities of recognition of clinical signs, of reporting or notification to the veterinarian or to the ICA, and the probability of sending samples in case of a suspicion of CSF (the questionnaires in Spanish are available upon request).

The surveillance at the slaughterhouse included only animals from commercial farms as stated above. Both fatteners and sows are sent to the slaughterhouse, so we assumed that only all

farrow-to-wean farms and farrow-to-finish farms with a census ≥ 100 sows sent batches of sows; while farrow-to-finish farms with less than 100 sows sent fatteners and culled animals together in the same batch. As pigs were sent in batches to the slaughterhouse, the surveillance unit was the batch and it was assumed that each batch was representative of the farm of origin. The elements that determined the probabilities of detection of an infected batch during antemortem and postmortem inspections are shown in Figure 2.

The sensitivity (Se_{farm} and $Se_{slaught}$) corresponds to the probability of CSFV detection on a farm *i* or in a slaughterhouse batch *i* by passive surveillance.

$$Se_{farm} = CS_i * (F_{c,vet} * F_{c,ICA}) * VetICA * VetSiS * SeElisa * SePCR \quad (5)$$

Where CS_i was the probability of manifestation of clinical signs, $F_{c,vet}$ and $F_{c,ICA}$ were the probabilities of a farmer calling a veterinary practitioner or the ICA on commercial or small farms, respectively, $VetSiS$ was the probability of a veterinarian suspecting CSF and $SeElisa$ and $SePCR$ were the sensitivity of the two laboratory tests.

$$Se_{slaught} = CS_i * (VetSiAMlorVetSiPMI) * VetICA_i * ICA_s * SeElisa * SePCR \quad (6)$$

Where $VetSiAM$ or $VetSiPMI$ were the probabilities of recognition of clinical signs by the veterinarian in antemortem or in postmortem inspection of fatteners or sows, $VetICA_i$ was the probability of the veterinary of slaughterhouse contacting the ICA after suspicion in sows or fatteners pigs and ICA_s was the probability of the ICA veterinarian taking samples.

The probability of detection of at least one infected farm by passive surveillance was given by Equations 7 and 8, where I corresponded to the total number of farms (Equation 7) and batches sent to slaughterhouse (Equation 8) under passive surveillance, $EPIH_i$ and $EPIH_c$ were the adjusted risk multiplied by design prevalence (P^*) according to the type of farm (in the case of slaughterhouse only for commercial farms) and Se_i corresponded to the probability of detection for farm *i* or slaughterhouse batch *i* (only commercial farms are included in the slaughterhouse surveillance).

$$Se_{farm} = 1 - \prod_{i=1}^I (1 - EPIH_i * Se_{farm}) \quad (7)$$

$$Se_{slaught} = 1 - \prod_{i=1}^I (1 - EPIH_c * Se_{slaught}) \quad (8)$$

2.6 | Sensitivity of the whole surveillance system

We assumed that active, passive on-farm and passive at slaughterhouse surveillances were independent components, the total sensitivity of the system for the detection of one CSFV-infected farm in zones 1 and 2 was calculated by the combination of the three probabilities (Equation 9) and can be defined as the probability of disease detection in at least one of the three components.

$$Se_{total} = 1 - [(1 - Se_{active}) * (1 - Se_{pas.farm}) * (1 - Se_{pas.slaughterhouse})] \quad (9)$$

The model was performed with R-project v3.6.3 using the library mc2d for Two-Dimensional Monte-Carlo Simulations running 5,000 iterations. This method reflected the variability in the risk and took into account the uncertainty associated with the risk estimate. The variability represented, for instance, temporal, geographical and/or individual heterogeneity of the risk for a given population. The uncertainty was understood as stemming from a lack of perfect knowledge about the Quantitative Risk Assessment model structure and associated parameters. In order to reflect the natural variability of a modelled risk, a Monte-Carlo simulation was used: the empirical distribution of the risk within the population was obtained from the mathematical combination of distributions reflecting the variability of parameters across the population. A two-dimensional (or second-order) Monte-Carlo simulation was proposed to superimpose the uncertainty in the risk estimates stemming from parameter uncertainty. (R Pouillot et al., 2016; Régis Pouillot & Delignette-Muller, 2010).

The values obtained from the questions to veterinarians were included in the model as inputs, extracting random samples by bootstrap resampling methods from discrete distributions (a short description of discrete distributions included in the Table 2 and exact values in Supporting Information).

The sensitivity of the surveillance for the design prevalences corresponding to 5 and 10 CSFV infected farms was calculated using a bootstrap method with the results of 1,000 iterations obtained randomly from the calculations for one farm.

3 | RESULTS

Total sensitivity of the surveillance system was similar in both zones. The probability of detecting a single infected farm was 28%–31% with a wide range (95%CI: 6%–55%), and the probability of detecting infection when 5 and 10 farms were infected was 83%–85% and 97%–98%, respectively. The passive surveillance component on farms showed the highest sensitivity for detection of a CSFV infected farm in both zones (22.8% in Zone 1 and 22.5% in Zone 2), the active surveillance and the passive surveillance at slaughterhouse presented values between 2.2% and 4.9% (Table 3).

The sensitivity of active surveillance for the detection of CSFV infection in zone 1 was slightly more sensitive in 2019 for all scenarios compared with 2018. Instead, in zone 2, the sensitivity of active surveillance was clearly higher in 2019 versus 2018 (3.3% and 2.2, respectively, for the detection of a single farm), due to the increase in the number of sampled farms (Table 3, Figure 3).

The animal level prevalence on commercial farms and the adjusted risk on commercial farms were the variables that showed the highest correlation with the obtained sensitivity of active surveillance on farms in both zones. (Spearman's rho statistic- tornado graphic included in the Supporting Information).

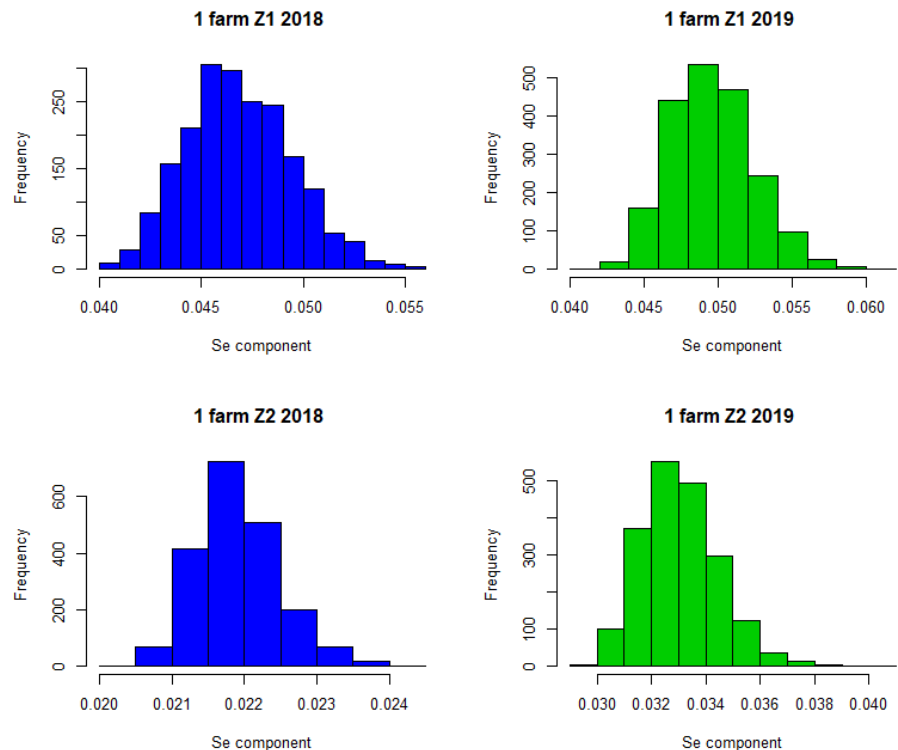
The sensitivity of passive surveillance for the detection of one farm (commercial or family) in Zone 1 was in a range between 0% and 48.5% (CI = 0.95) with a median value of 23%. For the detection of 5 farms, the median value was 73.4 (44.7%–88.3%). The probability of detection on a commercial farm was lower (6.2%) than on a family farm or on a backyard premise (17.8%); In Zone 2, sensitivity

TABLE 3 Sensitivity of each component and total surveillance in Zones 1 and 2 (median and in brackets 95% confidence interval)

Surveillance component	Zone 1			Zone 2		
	1 farm	5 farms	10 farms	1 farm	5 farms	10 farms
Active surveillance component						
Active (Design sampling 2018)	4.7 (4.2 – 5.2)	21.3 (19.4 – 23.6)	38.1 (35.1 – 41.7)	2.2 (2.1 – 2.3)	10.5 (10 – 11.1)	19.8 (19.1 – 20.9)
Active (Design sampling 2019)	4.9 (4.5 – 5.6)	22.4 (20.4 – 24.8)	39.7 (36.6 – 43.5)	3.3 (3.1 – 3.6)	15.4 (14.4 – 16.8)	28.4 (26.8 – 30.8)
Passive surveillance component on Farms						
Commercial	6.2 (0 – 12.7)	25 (12.1 – 37.1)	44.5 (27.5 – 55.5)	2.5 (0 – 7.4)	13.5 (5.9 – 21.9)	25.1 (15.1 – 35.2)
Backyard	17.8 (0 – 42.9)	64.4 (34.9 – 82.7)	87.5 (70.5 – 95.7)	19.4 (0 – 48.9)	70.2 (39.6 – 87.5)	91.2 (75.6 – 97.5)
Passive on farm	23.0 (0 – 48.5)	73.4 (44.7 – 88.3)	93.1 (79.1 – 97.8)	21.8 (0 – 51)	73.9 (47.9 – 89.3)	93.4 (81.5 – 98.2)
Passive surveillance component in Slaughterhouse						
Farrow-to-finish/ fattener farms	3.8 (0 – 13.5)	22.8 (7.2 – 38.7)	39.6 (18.2 – 54.8)	4.1 (0 – 7.7)	18.2 (8.5 – 26.3)	32.7 (21 – 42.7)
Farrow-to-wean farms ^a	1.2 (0 – 3.2)	6.4 (1.6 – 11.2)	12.3 (6 – 18.7)	0.4 (0 – 1)	2.2 (0.8 – 3.7)	4.5 (2.4 – 6.4)
Passive at slaughterhouse	5.3 (0 – 15.2)	27.3 (11.8 – 42.6)	47.2 (29.9 – 62.6)	4.5 (0 – 8.3)	20.2 (10.2 – 28.5)	35.6 (24.6 – 45.4)
All components						
Total	31.4 (7.2 – 54.1)	85.2 (67.3 – 93.7)	97.9 (93.4 – 99.3)	27.8 (6.4 – 55.1)	82.5 (65 – 92.9)	97 (91.3 – 99.2)

^aAlso includes farrow-to-finish farms with more than 100 sows (see text).

FIGURE 3 Distributions of the sensitivity of active surveillance in both areas for the detection of a single CSFV infected farm. Above: Zone 1, Below: Zone 2. Left: 2018 and Right: 2019



for the detection of a farm was similar to that observed in Zone 1 (21.8%); however, the differences between commercial and family farms were much wider (2.5% only on commercial farms and 19.4% on backyard/small farms) (Table 3 and Figure 4). The factors that showed most correlation with the efficiency of passive farm surveillance were as follows: if the farmer called the veterinary practitioner or ICA, if veterinary services suspected CSF from signs and took samples and if veterinary services suspected CSF after necropsy and took samples. (Spearman's rho statistic- tornado graphic included in the Supporting Information).

The passive slaughterhouse surveillance component had a sensitivity of 27.3% for the detection of five infected farms in Zone 1. It was much more sensitive to detecting the infection in fatteners or in sows coming from farrow-to-finish farms (22.8%) than in sows (6.4%). In zone 2, the sensitivity of this component was lower (20.2%), and it showed a similar pattern for the detection of fatteners and sows (Table 3). Figure 5 shows the distribution of probabilities for a single infected farm in Zone 1.

The analysis of sensitivity of the model showed that the variables Veterinarian contacting ICA after suspicions and Veterinarian recognizing clinical signs in ante-mortem and post-mortem inspection had the highest correlation with passive surveillance efficiency. (Spearman's rho statistic- tornado graphic included in the Supporting Information).

4 | DISCUSSION

CSF was endemic in Colombia until 2007 when the disease was eradicated. In 2013, the disease was reintroduced. Until 2018, 134

outbreaks had been reported in different departments of the Atlantic Coast region, Norte de Santander and Santander of which 84% were detected by passive surveillance (Pineda et al., 2020). The endemic situation of CSF in the Atlantic Coast departments as well as in neighbouring countries supposes a threat for the regions that remain free of the infection and where the animals are not vaccinated. If infection spreads in these zones, early detection of outbreaks is of paramount importance to achieve rapid and less expensive elimination of the virus on affected farms. According to our results, the sensitivity of the surveillance in the two studied zones is high. The probability of detection of a single affected farm was about 30% and the probability of disease detection when 10 premises were affected was more than 97%.

On-farm passive surveillance was by far the most sensitive component, with a median sensitivity to detect a single farm of 23%, which increased to 93% when 10 premises were infected. The sensitivity of the surveillance in the studied zones was much higher on backyard and small farms than on commercial ones, despite the capacity and predisposition to declare infections on these premises is supposed to be low (Marta Hernández-Jover et al., 2019; Limon et al., 2014). This high sensitivity can be attributed to the Colombian strong voluntary sentinel network made up of more than 5,000 members trained in diseases under official control. The network was responsible for the detection of 21 of the 134 CSF outbreaks that occurred in the endemic region (Pineda et al., 2020). This sentinel network is particularly focused on backyard premises due to their higher risk of infection and lower application of biosecurity measures. Between 2018 and 2020, 85 CSF suspicions were reported in Zone 1 and 198 in Zone 2 (ICA, 2020) which reflect good reporting in the free zones. Another reason for this high sensitivity in small farms would be

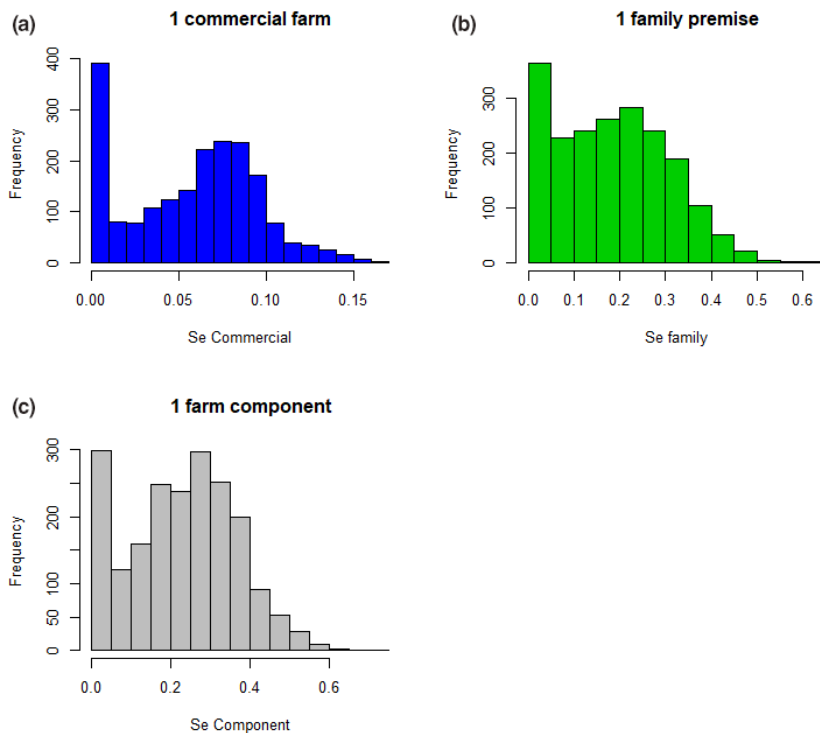


FIGURE 4 Distributions of the sensitivity of passive surveillance in Zone 1 for the detection of: (a) one commercial farm, (b) one family premise or (c) any type of farm

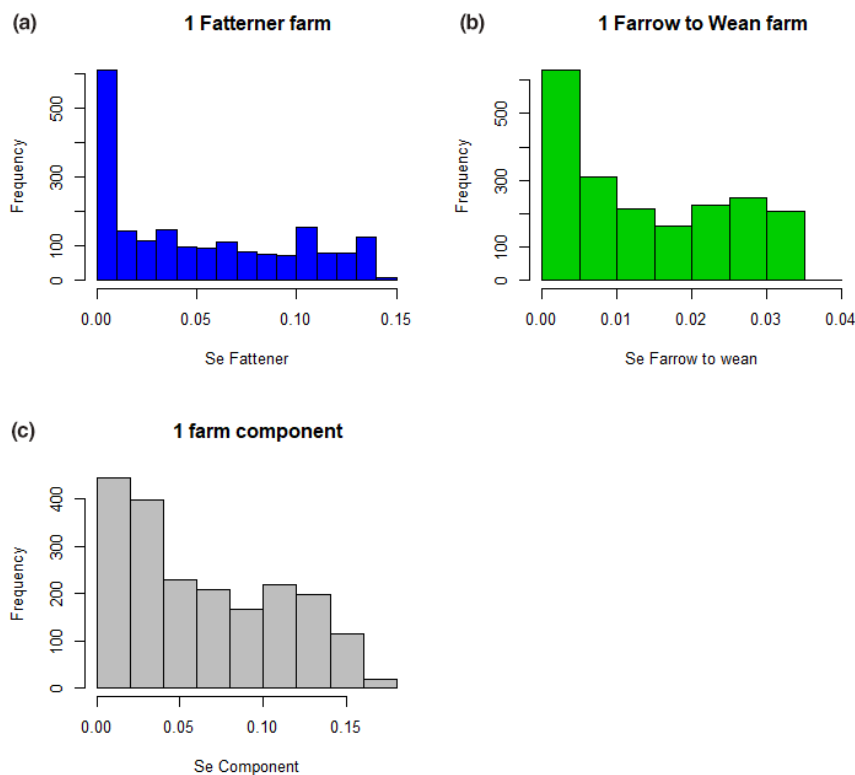


FIGURE 5 Distributions of passive surveillance sensitivity at slaughterhouse for the detection of one: (a) farm lot Farrow-to-finish / fattener, (b) farrow-to-wean or (c) any type of farm in Zone 1

that low-income farmers directly contact ICA services if their animals become diseased because they are free of cost.

On the contrary, we can assume an overestimation of the sensitivity due to bias related to backyard producers. The main reasons for this are, that only a part of them are registered in the official services while the rest probably are more reluctant to report the

disease; backyard producers get used to treat the animals themselves and report only when they do not succeed (Marta Hernández-Jover et al., 2019); and, finally, there are also farmers who do not report to avoid quarantine and restrictions (Elbers et al., 2010).

Commercial farms have the advantage of having more experienced and trained farmers, veterinarians and pig handlers, however,

they report less. Therefore, there needs to be a strategy to encourage the reporting of suspicions, since that contact with a veterinary practitioner or ICA is one of the factors that most affect the sensitivity of passive surveillance.

Specialization of veterinarians and improving awareness of clinical signs along with pathological and virological examination for atypical signs are effective methods for the early identification of a CSF outbreak (Crauwels et al., 1999; Engel et al., 2005). Veterinarians are considered to be a trusted stakeholder and, therefore, they can have a strong influence on the behaviour of smallholders improving the willingness of disease reporting (Hayes et al., 2017; M. Hernández-Jover et al., 2012). Also, the relationship between producers and veterinarians play an important role in surveillance. To achieve effective passive surveillance on the part of the producer, they must trust the person receiving or providing the information. The reporting of suspected diseases by the producer could be improved by increasing their awareness of the disease seriousness, increasing the frequency and extent of animal inspections by the owners, and improving their willingness to investigate problems (Garner et al., 2016). Therefore, this is where the education and awareness programs play an important role in passive surveillance.

There is a need to maintain education and training directed at ICA staff and especially at veterinarians and other professionals working on commercial farms in order not to ignore the latent risk posed by the endemic situation on the Atlantic Coast.

Passive surveillance at slaughterhouses had a median sensitivity for the detection of a single infected farm of 5% in both Zones. It showed higher sensitivity for detecting farrow-to-finish or fattener farms, because of the higher volume of pigs sent to slaughterhouse compared with farrow-to-wean farms that occasionally send sows or boars. However, it is important to consider that in Colombia, only commercial farms send animals to slaughterhouses because most backyard premises slaughter the animals on their own facilities.

In recent years, no CSF suspicions have been reported by slaughterhouse veterinarians in Colombia, probably due to the low number of outbreaks on commercial farms and also to the low awareness of those veterinarians. The National Drug and Food Surveillance Institute (INVIMA) has a regular training and disease detection manuals for slaughterhouse veterinarians and the ICA and Porkcolombia conducts occasional training for them, but it should be reinforced in order to improve the recognition of clinical signs and lesions and knowledge about the procedure in case of a CSF suspicion.

The sensitivity of active surveillance ranged from 2.2% to 4.9% (depending on the zone and year) for detecting the infection if a single farm was infected. The differences between both zones can be attributed to the differences in the design prevalence: 1 farm supposed a higher prevalence rate in Zone 1 ($7.4 \cdot 10^{-5}$) due to the lower number of farms compared with Zone 2 ($4.9 \cdot 10^{-5}$).

In Colombia, active surveillance is based on the combination of the Elisa test and PCR. According to (Panyasing et al., 2018), this is the best option for effective CSFV surveillance, especially in the case of low virulence strains in order to detect both early and old infections. Other active surveillance activities carried out in the

country, such as inspection at collection points, trade fairs and some premises and small farms were not considered in the study.

One of the reasons for the higher sensitivity of passive surveillance is that it is performed on a significant part of the farms in the country while only animals from commercial farms are examined at slaughterhouse and only a sample of farms is tested by active surveillance. Furthermore, passive surveillance on farms is based on the continuous monitoring of animals by farmers and veterinarians allowing the daily detection of diseased animals while the other components are evaluated only in a given period of the year when farms are monitored by active surveillance or when a batch of animals is sent to the slaughterhouse. Reporting by farmers depends on awareness that it is a serious disease (Garner et al., 2016), recognition of clinical signs (Elbers et al., 2010), and trust in their veterinarian or contact personnel (Hayes et al., 2017) and also on the transparency of the reporting process to reduce the uncertainty of the farmers (Elbers et al., 2010). Significant efforts have been done by the veterinary authorities and by Porkcolombia to overcome these problems.

On the other hand, the advantage of active surveillance is its ability to detect subclinical animals or animals infected with low virulent strains, in which clinical signs may go unnoticed. Consequently, all components of the surveillance system are important and complementary to detect infection in the case of an outbreak of CSF.

The stochastic scenario tree model methodology developed by Martin, Cameron, Barfod, et al., (2007) is useful not only for determining the system's efficiency but also for the identification of the components that could be improved for earlier detection of diseases.

This method can provide information about the risk of introduction of exotic disease and has the advantage of being able to use surveillance data from multiple sources (Christensen & Vallières, 2016). It also enables evaluation of possible ways to improve sensitivity as has been described for different diseases: increasing awareness among veterinarians and farmers could improve the probability of detection of Avian Influenza in Catalonia (Spain) (Alba et al., 2010); and increasing disease awareness among saleyard and abattoir stockmen and increasing the presence of inspectors and identifying herds with a FMD higher risk can improve early detection capacity of emerging diseases in Australia (M. Hernández-Jover et al., 2011), while another example could be awareness among hunters, which would increase passive surveillance sensitivity for bovine tuberculosis in wildlife (Rivière et al., 2015).

This study was carried out with information from the surveillance activities applied in the country. Information related to probabilities of farmers and veterinarians suspecting CSFV infection and declaring the disease was obtained from surveys aimed at veterinarians; therefore, it can reflect the current knowledge and attitudes related to CSF surveillance in the two zones. Among the limitations of this study are the parameters obtained with questionnaires (included in the Supporting Information) in which some clinical cases were used as examples. Alternative cases with less or more evident clinical signs would probably have modified the responses. Also the answers were probably conditioned by the expected response rather than by their actual attitude in the case of a real situation that could go

unnoticed or they might even be reluctant to report the suspicion, which may have led to an overestimation of passive surveillance sensitivity. The results present variability due to the different perception by the veterinarians and the differences between the cases included in the questionnaires. Uncertainty is also assumed in the diagnostic test due to possible differences in the results between laboratories. Therefore, the sensitivity of the farms and slaughterhouse contains point estimates along with their credibility intervals in the dimensions of variability and uncertainty.

Other factors have been previously identified as affecting the reporting of disease outbreaks: the first is the lack of knowledge on the early signs of CSF evidenced in areas where the disease has not occurred in the last 10 years and also negative opinions regarding control measures, dissatisfaction with post-reporting procedures, lack of trust in government bodies, uncertainty and lack of transparency of reporting procedures (Elbers et al., 2010). This situation may be especially common on commercial farms due to problems with the financial compensation for destroyed animals and the commercial repercussions that in many cases can lead to the definitive closure of the farm.

In conclusion, the sensitivity of the surveillance in the two studied zones was quite high, passive surveillance at farm level being higher than detection at slaughterhouse and active surveillance. Furthermore, it is important to strengthen factors that have an effect on the system's sensitivity, such as the capacity of farmers and farm and slaughterhouse veterinarians to detect clinical signs of CSF and to educate on the importance of reporting.

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CONFLICT OF INTEREST

The authors declare no competing interest. Co-authors working at the Colombian Association of Pig Farmers (PorkColombia) helped to collect and coordinate data but had no role in its analysis and interpretation.

ETHICS STATEMENT

This study did not need any ethical approval, most data are public and the surveys applied to veterinarians only collected their opinions, which were included in the models as probability values.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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